IEEE BIBM 2026 — Doctoral Forum Cover Note

Dear Dr. Yi He and Dr. Xingquan Zhu,

The attached manuscript, "Binned Spiking Reservoir Encoding for Perturbation-Characterized Affective EEG Signal Analysis," evaluates a fixed leaky integrate-and-fire (LIF) reservoir with a six-bin spike-count (BSC6) code as a nonlinear temporal encoding operator for affective EEG/ERP observations. Using subject-grouped validation against a classical bandpower/ERP baseline, the study reports balanced accuracy, macro-F1, and macro one-vs-rest AUC, and characterizes the encoding under additive amplitude noise, temporal jitter, and channel dropout, separating representation-level corruption from full raw-signal recomputation.

The result is deliberately bounded. The reservoir embedding is not a uniformly stronger static classifier than classical ERP-derived features, but it exhibits a distinct perturbation-response profile: it retains balanced accuracy more strongly under representation-level amplitude noise and channel dropout, while remaining sensitive to temporal misalignment and severe channel loss when the representation is recomputed from corrupted signals. Characterizing what temporal information a spiking representation preserves, transforms, or suppresses—rather than reporting a single clean-accuracy number—gives a reproducible operating-regime account. By characterizing how a temporal reservoir representation changes under amplitude noise, temporal jitter, and channel dropout, the study supports more auditable biomedical signal-analysis pipelines for EEG/ERP data before downstream clinical interpretation is considered.

We believe the work fits the IEEE BIBM Doctoral Forum. Its contribution is original and relevant to the BIBM community: it reframes a spiking reservoir as a perturbation-characterized temporal measurement operator for affective EEG/ERP rather than as another endpoint classifier. The methodology and evaluation are designed for soundness and clarity—subject-grouped validation, a classical EEG/ERP baseline, balanced accuracy, macro-F1, and macro one-vs-rest AUC, with a fold-local PCA sensitivity analysis that bounds the transductive-projection concern—and the problem it addresses, representational stability of biomedical signals under controlled corruption, is a significant and underreported question for reproducible EEG/ERP analysis. As student-led doctoral-stage research, the first author is the student applicant, primary contributor, and intended presenter; eligibility status (Ph.D. candidate/recently defended doctoral researcher in Electrical and Computer Engineering at Stony Brook University) will be stated accurately at submission, and a one-page CV is appended as requested in the call. This submission forms one component of a broader doctoral framework for neuromorphic signal processing of affective neural data and is intentionally focused on the reservoir temporal-encoding component; it does not duplicate the broader journal manuscript under preparation.

Sincerely,

Andrew A. Lane